

Jonathan Barnes | Analytics Engineer and Data Scientist

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I solve chaotic data problems from the infrastructure up, combining systems engineering with statistical modeling to build, production-grade platforms that turn unstable systems into reliable, analysis-ready foundations.

PROFESSIONAL EXPERIENCE

University of Minnesota, Minneapolis, MN

Analytics Engineer

(May 2025 - Present)

- Established a production analytical data platform for all HHS awards, consolidating 400 inconsistent source fields into 4M records and 26 standardized variables, enabling global funding analysis that was previously infeasible at scale.
- Implemented a reproducible transformation layer to normalize heterogeneous federal data and enforce canonical variable definitions, including multi-field coalescing, deterministic tie-breaking to prevent aggregate duplication, null-control logic, schema drift handling, and reference-backed validation to produce stable, analysis-ready variables.
- Engineered a repeatable, memory-efficient normalization pipeline in Polars processing heterogeneous federal data end-to-end in under 2 minutes using under 12GB RAM, reducing memory usage by 80%+ and avoiding high-memory or distributed infrastructure.
- Designed a fault-tolerant ingestion and orchestration layer backed by DuckDB that replaced direct dependence on unreliable, rate-limited government APIs with a persistent analytical store for ad-hoc analysis.
- Built a governed self-serve interface enabling non-technical stakeholders to trigger refreshes, monitor pipeline health, and execute constrained ad-hoc queries against the persistent store, reducing routine engineering intervention.
- Built a custom Polars-to-Tableau Hyper export layer to reduce analytic file size by over 90%, significantly improving enterprise dashboard performance while preserving full granularity.

Graduate Teaching Assistant - Biostatistics

(Aug 2024 - May 2025)

- Taught and evaluated applied statistical analysis in R, improving student code quality and statistical reasoning.

Graduate Research Assistant

(Oct 2023 - May 2024)

- Designed custom plasma biomarker phenotypes and gene expression features to enable survival modeling.
- Applied Cox proportional hazards regression to model time to cognitive decline using biomarker phenotypes.

University of St. Thomas, St. Paul and Minneapolis, MN

ITS Support Technician Intern & Senior Student Technician

(Aug 2021 - Aug 2023)

- Designed and deployed a Python-based internal platform that standardized high-variance service desk workflows, replacing a legacy tool and becoming the primary interface used by technicians during live support.
- Built guarded, technician-ready interfaces over operating system, directory, and network queries that return consistent diagnostic and identity information and provide access to administrative systems without requiring command-line use.
- Resolved or consulted on 4,500+ support tickets across hardware, software, networking, and security, serving as the senior escalation layer between the service desk and support teams while shaping the design of standardized processes.
- Mentored junior student technicians to independently handle complex service desk operations and consultations.
- Imaged and administered software on 1,500+ replacement computers and leveraged retired hardware to develop an in-house virtual lab replacing AWS EC2 instances and reducing infrastructure costs.
- ITS Outstanding Student Award.

EDUCATION

University of Minnesota School of Public Health, Minneapolis, MN

Master of Public Health - Public Health Data Science, GPA 3.7/4.0

Feb 2026

- Capstone – Data Platform Evaluating Federal Spending and Systems.
- President's Student Leadership & Service Award Nominee (2025)

University of St. Thomas, St. Paul, MN

Bachelor of Science - Data Analytics, completed while employed full-time

- Minors - Biology (Biochemistry & Computational Biology courses), Applied Statistics.
- Capstone - Development of Obesity and Nutritional Trends Over Time in the United States Using Combined NHANES.

SKILLS

Languages: Python, R, SQL, Bash.

Tools: Git, CI/CD, GitHub Actions, Docker, Kubernetes, Kafka, Spark, dbt, ArcGIS.

Competencies: ETL Pipelines, Data Modeling, System Design, Feature Engineering, Data Acquisition, Longitudinal Data Analysis, Data Visualization, Machine Learning, NLP, LLMs, Experimental Design, Health Informatics, EHR, HIPAA.

Modeling Methods: Regression, Regularization, Tree Models, Supervised Learning, Unsupervised Learning, Monte Carlo Simulation, Dimensionality Reduction, Imputation, Mixed Effects Models, Survival Analysis, Structural Equation Modeling.

Projects

Public Sector Job Data Platform

- Architecting a public health workforce intelligence platform that ingests and normalizes job postings from 1,700+ government health department websites using a platform-aware ATS classification system and targeted scraping strategies.
- Building a Kafka-backed producer-consumer pipeline to decouple scraping, LLM-assisted HTML parsing, and structured data extraction, deployed across a self-hosted Kubernetes cluster with distributed LLM inference.
- Designing a search interface to improve access to public health job postings, giving equal visibility to small rural health departments and large state agencies for workforce research.

Technical Skills: Python, Kafka, dbt, Spark, PostgreSQL, Kubernetes, LLMs, Web Scraping, Data Architecture, Distributed Systems

Self-Hosted AI & Data Infrastructure Compute Cluster

- Personally built and operated a self-hosted Kubernetes (k3s) cluster on repurposed 2013 Mac Pro and gaming laptop hardware, running 30+ containerized services 24/7 with zero cloud infrastructure spend and minimized resource waste.
- Deployed and load-balanced LLM inference across dual AMD FirePro D700 GPUs (12GB VRAM) using Ollama, serving models locally as an embedded pipeline processing layer and eliminating cloud API costs entirely.
- Developing webhook-driven automation workflows using local LLM inference and Bayesian models to prioritize tasks, optimize schedules, and manage resource/energy constraints across self-hosted services.

Technical Skills: Linux, Docker, Kubernetes (k3s), Ollama, Traefik, Python, RAG, Vector Databases, DevOps, Systems Architecture.

Public Health Federal Funding Evaluation Platform & Dashboard (Grant Supported)

- Assembled a multi-million record health funding dataset across prime awards, sub-awards, and derived transactions, using identity resolution to reconcile inconsistent award and recipient relationships that prevented reliable large-scale analysis.
- [Project Page](#)

Technical Skills: Python, Polars, SQL, DuckDB, Apache Parquet, Tableau (Hyper), Data Modeling, Schema Design, Entity Resolution, Data Pipelines, Data Warehousing, Data Infrastructure Design.

Tech Desk Tools Python

- Implemented a user-facing Python application that unified Tech Desk identity lookup, diagnostics, and institutional tool access, using identity derivation and privilege-aware automation across OS and directory services to produce stable technician-ready outputs and ensure consistent, secure behavior across varied account and permission states.
- Adopted as the standard Tech Desk application, supporting continuous daily operations since early 2023.
- [Tool Overview](#)

Technical Skills: Python, PowerShell, Bash, App Development, Active Directory, Azure, Systems Automation, Workflow Orchestration, Error Handling & Defensive Programming, Technical Writing, Packaging and Deployment.

Mapping Misery – Predicting Poor Physical and Mental Health Days in the U.S.

- Built county-level predictive models of mean-centered poor physical and mental health days across ~3,200 U.S. counties; engineered longitudinal features by encoding 10-year trajectories across 15 indicators into ordered levels (treating missingness as signal), learned latent structure via MCA, and derived cluster-level patterns using Gaussian Mixture Models.
- Trained elastic net models with 25-fold cross-validated regularization and CART-based imputation; achieved $R^2 = 0.844$ (RMSE = 0.263, MAE = 0.208) for physical health and $R^2 = 0.645$ (RMSE = 0.375, MAE = 0.295) for mental health.
- [Project Page](#)

Technical Skills: Feature Engineering, Longitudinal Data Analysis, Clustering, Elastic Net Regression, Cross-Validation, Imputation, R.

Predicting Child Mortality and Cesarean Delivery Outcomes with Generalized Linear Mixed Models

- Co-developed binomial mixed-effects logistic regression models with community-level random intercepts (n=1,897; 60 communities) to assess predictors of child mortality and C-section; found maternal age increased odds of child death ~5% per year (p=0.001) and that hospital delivery and insurance strongly predicted C-section (as expected).
- [Project Page](#)

Skills & Tools: Mixed Effects Modeling, Multilevel Modeling, Clustered Data Analysis, Survey Data Analysis, Sensitivity Analysis.

Applied Biostatistics Introduction Modules: Shiny Suite (Grant Funded)

- Designed applied biostatistics modules translating into intuitive frameworks with supporting GitHub hosted Shiny applications.
- [Shiny App - Intuition Lab: Odds, Base Rates Fallacy, Ratio Changes, Gambler's Fallacy](#)
- [Shiny App - Intuition Lab: Sampling, CLT, Regression to the Mean, Spread and Confidence](#)
- [Shiny App - Intuition Lab: Effect Size, Sample Size, Power, Trade-Offs](#)

Technical Skills: Statistical Reasoning, Probability, Curriculum Design, Technical Writing, R, Data Visualization, JavaScript.